

SnapNet

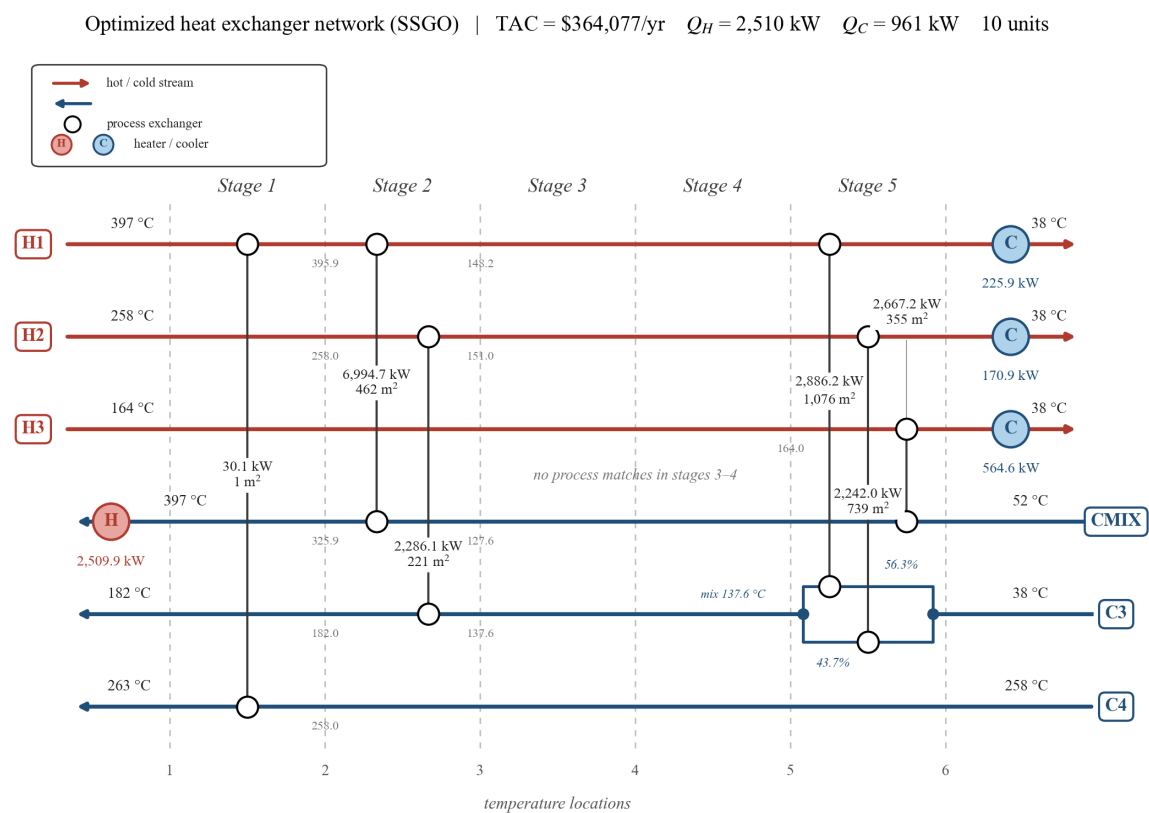
HEN Network Diagrams from SnapOpt Solutions — User Guide

User Guide

Version 1.0 • July 2026 • SnapTera suite

Created by Mohammed Ali Mansoury

SnapTera: From Targeting to Optimization to Network Design



SnapNet turns a SnapOpt solution into the final deliverable: a publication-ready stagewise grid diagram of the optimized heat exchanger network — stream lines, every exchanger with its duty and area, coolers and heaters, stage-boundary temperatures, and automatic stream splits with duty fractions and mixing temperatures.

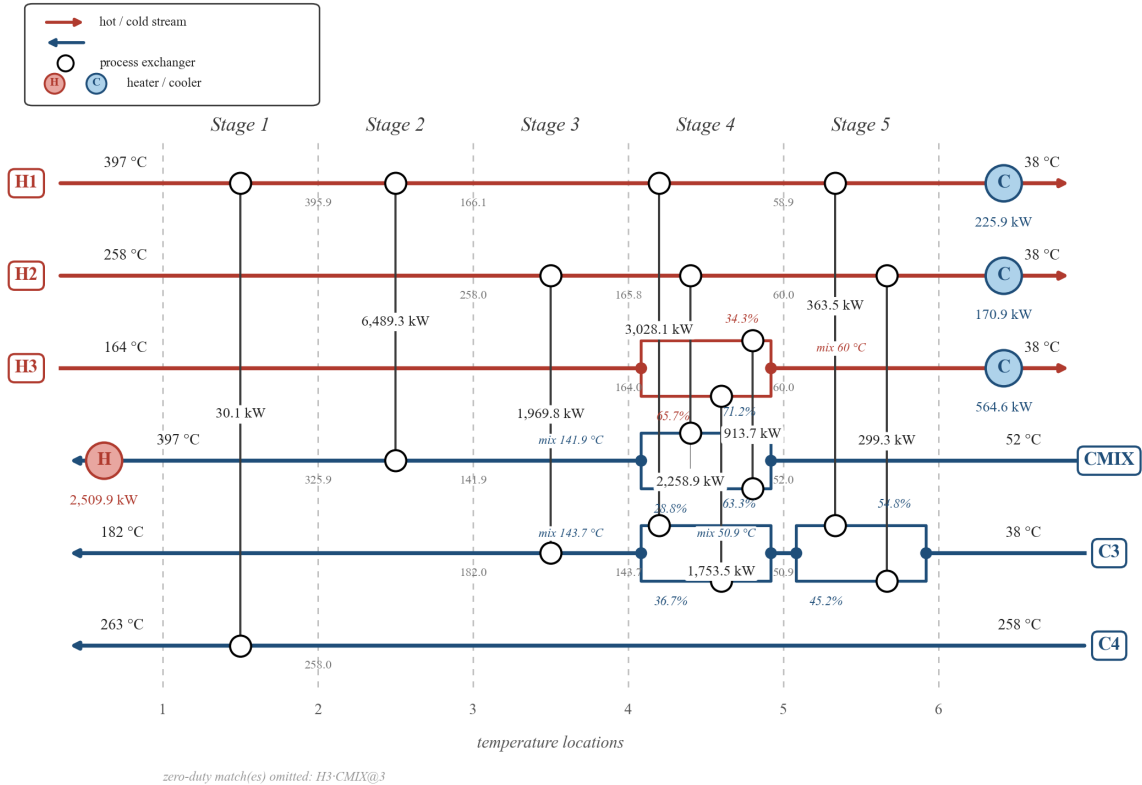
1. Requirements and quick start

SnapNet is a single Python script (`snapnet_draw.py`, needs Python 3 with matplotlib). Its input is the `solution.json` that SnapOpt writes into every run's `results\` folder.

Command	Effect
<code>python snapnet_draw.py</code>	Draw the default solution (SnapOpt\results\solution.json) to <code>hen_grid.png</code> next to it.
<code>python snapnet_draw.py run\results\solution.json</code>	Draw a specific SnapOpt run.
<code>... -o net.png --areas</code>	Choose the output file; print exchanger areas under the duties.
<code>... --svg --pdf --dpi 600</code>	Also export SVG and PDF; raise the PNG resolution.
<code>... --streams input.json</code>	Use exact stream data (auto-detected next to the solution when present).
<code>... --no-temps --title "My network"</code>	Hide stage-boundary temperatures; custom title.

2. Reading the diagram

Element	Meaning
Red / blue arrows	Hot streams flow left→right, cold streams right→left; supply and target temperatures at the line ends.
White circles + connector	A process exchanger between the two streams, with its duty (kW) and, with <code>--areas</code> , its area (m ²). A thin gray leader line ties a label to its exchanger wherever space was tight.
H / C circles	Fired heater (on cold streams, left) and cooler (on hot streams, right) with utility duties.
Dashed grid, Stage k	The superstructure stages and temperature locations; small gray numbers under each line are the stream's stage-boundary temperatures.
Forked line	A stream SPLIT: parallel branches with duty fractions (%), re-mixed isothermally at the temperature shown after the merge node.
Footnote	Zero-duty matches (solver bookkeeping artifacts) are omitted and listed.



A dense diagnostic case (DICOPT local incumbent): four simultaneous splits in stages 4–5, including a hot-side split of H3 — drawn automatically.

3. Where the stream data comes from

The diagram needs each stream's CP and supply/target temperatures. SnapNet looks for the run's `input.json` next to the solution (written by every SnapOpt run) and uses those exact values. When there is none — e.g. a solution imported straight from a GAMS listing — SnapNet derives the data from the solution itself via stage energy balances: CP from the largest-duty stage (duty / temperature change), targets from the utility duties. On the thesis case the derived CPs match the true values to $\sim 3 \cdot 10^{-4}$ relative.

4. Validation

The automated test (`test_snapnet_draw.py`) checks the drawn network against the hand-built thesis figure: the C3 split in stage 5 at 56.3 / 43.7 % with isothermal mixing at 137.6 °C, all six matches and duties, the derived stream table, and — on the DICOPT case — all four splits and the omitted zero-duty match.

5. Licence

SnapNet is proprietary software, licensed — not sold. Subject to payment of the applicable licence fee, you are granted a personal, non-exclusive, non-transferable licence to install and use it for your own internal purposes, granted per named user: you may install it on any number of computers you alone control, but the licence may not be shared, pooled or used concurrently by more than one person. Until a licence fee has been paid you may use it only to evaluate it — privately, non-commercially, and not for any result that informs a real design, purchase, tender, report or deliverable.

You may not copy, distribute, publish, sublicense, rent, lease, lend or sell the software; modify it or create derivative works; reverse engineer, decompile or unlock its source code; remove or alter any copyright, licence or attribution notice; or use it to build, train or improve a competing product.

No warranty. The software is provided “as is”, without warranty of any kind, express or implied. Engineering results are indicative, not a design: every number must be independently checked by a qualified engineer before it is relied on for any design, procurement, safety, operational or financial decision. Liability is limited to the licence fee actually paid.

Microsoft Excel, Python, GAMS and the GAMS solvers are not included and must be licensed separately. Full agreement: SnapTera/EULA.md. Contact: atsnaptera@gmail.com

SnapNet v1.0 — created by Mohammed Ali Mansoury, 2026. Licensed per named user on purchase; see the End User Licence Agreement above. Part of the SnapTera suite with SnapTarget (targeting) and SnapOpt (optimization). Contact: atsnaptera@gmail.com